

Lifelong Opportunity Guide 78 — Web and Digital Interface Designer

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U.S. occupation benchmark: O*NET-SOC 15-1255.00 — Web and Digital Interface Designers

Canada comparison: NOC 21233 — Web designers

Colombia comparison: CUOC 21662 — Diseñadores de imágenes computarizadas e interacción digital

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What this career is

Web and digital interface designers help make websites, applications, services, dashboards, portals and other digital experiences understandable, usable and visually coherent. They may research user needs, organize information, sketch flows, create wireframes and prototypes, define interface patterns, prepare visual specifications, test designs, review accessibility considerations and collaborate with developers through implementation.

The work sits between visual communication, interaction design, product thinking and technology. Job titles vary widely. Employers may use titles such as web designer, UI designer, UX/UI designer, interaction designer, digital designer, product designer, experience designer, web design specialist, visual interface designer or front-end designer. Those titles are not automatically equivalent. Always read the actual responsibilities, required skills and authority in the job posting.

This guide uses **O*NET-SOC 15-1255.00 — Web and Digital Interface Designers** as its primary U.S. benchmark. O*NET describes work that includes designing websites or digital user interfaces; developing and testing layouts, interfaces, functionality and navigation for compatibility and usability; evaluating design against web and accessibility standards; analyzing usage; and supporting human-computer interaction. Some jobs also include HTML, CSS, JavaScript or other code, while others are primarily design roles.

Why this role can be an opportunity

Digital interfaces are used in nearly every sector: business, government, health care, education, banking, commerce, transportation, hospitality, media, nonprofit services and internal enterprise systems. The occupation

can suit people who enjoy combining visual judgment with problem solving and structured thinking.

Transferable experience may come from:

- graphic design;
- customer service;
- marketing and communications;
- web content management;
- technical support;
- software testing;
- front-end development;
- business analysis;
- research;
- accessibility work;
- education;
- documentation;
- project coordination;
- photography or digital media.

A strong entry path is usually portfolio-driven. A learner does not need to buy every commercial tool before testing career fit. Free or low-cost tools, public design systems, supervised projects, structured practice briefs and accessible prototypes can show whether the work fits before major spending.

Role boundaries: UI, UX, web design, front-end development and product management

Employers use these terms differently, but the distinctions below are useful.

UI or visual interface design

Typical emphasis:

- layout and hierarchy;
- typography and color;
- components and patterns;
- spacing and design tokens;
- states such as hover, focus, selected, error and disabled;
- responsive layouts;
- visual consistency;
- design-system use.

UX and interaction design

Typical emphasis:

- user goals and tasks;
- research and feedback;
- information architecture;
- task flows;
- navigation;
- wireframes;
- prototypes;
- usability testing;
- interaction behavior;
- reducing confusion and unnecessary effort.

Web design

Typical emphasis may combine:

- visual interface design;
- content layout;
- responsive page structure;
- navigation;
- accessibility awareness;
- basic search/analytics considerations;
- content-management constraints;
- some HTML/CSS or no-code/low-code implementation, depending on the employer.

Front-end development

Typical emphasis:

- implementing interfaces in code;
- HTML, CSS, JavaScript/TypeScript and frameworks;
- component behavior;
- browser compatibility;
- performance;
- integration with services/APIs;
- automated testing;
- production engineering practices.

A designer may know front-end code without being the production software engineer. A front-end developer may contribute design judgment without owning the product's UX strategy.

Product management

Typical emphasis:

- product strategy;
- priorities and roadmap;
- business outcomes;
- stakeholder alignment;
- requirements and tradeoffs;
- release decisions;
- product metrics.

A designer may inform these decisions but should not claim product-management authority unless it is actually assigned.

Typical responsibilities

Depending on the organization and level, a web and digital interface designer may:

- clarify the user, business and service problem;
- review existing research, analytics, support tickets or usability findings;
- help plan user research under approved procedures;
- create personas or journey maps when useful and evidence-based;
- organize content into an information architecture;
- create sitemaps and navigation models;
- map user flows and task states;
- create low-, medium- or high-fidelity wireframes and prototypes;
- design responsive layouts for desktop, tablet and mobile contexts;
- define component states and interaction behavior;
- apply approved brand and design-system rules;
- document spacing, typography, color and component usage;
- prepare assets and specifications for developers;
- collaborate with front-end and back-end developers;
- review implementation against approved designs;
- participate in usability testing and document findings;
- identify accessibility risks and coordinate remediation with the responsible team;
- review content clarity and form instructions;
- design loading, empty, validation, error and success states;
- support localization and internationalization planning;
- protect research participants and confidential product information;
- maintain version history and approval records;

- update designs after evidence, testing or stakeholder review.

Authority and professional boundaries

A web or interface design title does **not** automatically authorize a worker to:

- deploy production code;
- approve a production release;
- change security controls;
- create or approve privacy policy;
- make legal claims about accessibility or regulatory compliance;
- certify WCAG conformance from mock-ups alone;
- collect sensitive user data without approved research/privacy procedures;
- install unapproved analytics or tracking technologies;
- change authentication, authorization or payment logic independently;
- upload confidential code, user research, credentials, customer data or product plans into an unapproved AI service;
- promise that a design is secure, compliant, unbiased or accessible simply because it follows a checklist;
- represent generated or borrowed design work as original human work when disclosure or licensing rules require otherwise.

When the issue is outside design authority, escalate to the appropriate product owner, engineering team, security team, privacy/legal function, accessibility specialist, content owner, research lead, data-protection function or other authorized decision maker.

The interface-design lifecycle

A repeatable process reduces rework and makes design decisions easier to explain.

1. Understand

Clarify:

- who the intended users are;
- what they need to accomplish;
- what business or service outcome matters;
- what evidence already exists;
- technical constraints;
- content constraints;
- accessibility requirements;
- privacy/security constraints;

- supported devices and environments;
- decision owners.

Do not invent a user problem simply to justify a preferred design.

2. Structure

Create or refine:

- information architecture;
- page hierarchy;
- content grouping;
- navigation;
- user flows;
- form sequences;
- decision points;
- error and recovery paths.

3. Prototype

Move from lower fidelity to higher fidelity only as needed. A prototype can test structure and behavior before expensive implementation.

Include important states such as:

- first use;
- empty state;
- loading;
- validation;
- error;
- success;
- permissions restrictions;
- offline/connection problem where relevant;
- keyboard focus;
- responsive behavior.

4. Test and review

Use appropriate evidence:

- usability sessions;
- stakeholder review;
- analytics;
- accessibility review;
- content review;
- technical feasibility review;
- security/privacy review where relevant;

- implementation review.

One opinion is not automatically research. Record what was observed, what changed and what remains uncertain.

5. Handoff and implementation support

Provide developers with enough information to reproduce the intended behavior:

- layouts and breakpoints;
- states;
- component rules;
- assets;
- design tokens where used;
- content and labels;
- accessibility expectations;
- interaction notes;
- acceptance criteria or test notes when assigned.

6. Learn after release

If the organization measures outcomes, compare real use with the original goals. Treat design as an evidence-informed lifecycle rather than a one-time picture.

Core design skills

Visual design

- hierarchy;
- typography;
- spacing;
- grids;
- color;
- iconography;
- imagery;
- responsive layout;
- consistency;
- visual states.

Interaction design

- task flows;
- navigation;
- controls;
- feedback;
- progressive disclosure;

- error prevention and recovery;
- predictable behavior;
- keyboard/focus behavior;
- touch and pointer considerations.

Information architecture and content

- grouping and labeling;
- navigation structure;
- search/filter concepts;
- concise interface writing;
- form labels and instructions;
- meaningful headings;
- content priority;
- localization-aware layouts.

Research and evaluation

- asking neutral questions;
- observing task behavior;
- separating evidence from assumptions;
- documenting findings;
- prioritizing issues;
- recognizing sample limitations;
- protecting participant data;
- knowing when specialist research support is required.

Technology

Tools change rapidly. Employers may use software for:

- interface design and prototyping;
- whiteboarding;
- design systems;
- image/vector editing;
- project tracking;
- user research;
- analytics;
- testing;
- documentation;
- version control;
- content management;
- front-end development;
- approved AI assistance.

Current U.S. employer-posting data linked by O*NET show strong demand for **Figma**, with CSS, JavaScript, HTML, React and other tools also appearing in postings. Those postings show market demand, not mandatory skills for every design job.

Learn transferable concepts: components, variants, tokens, auto layout, responsive constraints, styles, interaction states, version history, annotations, assets and developer handoff. A product name can change; the design logic remains useful.

Accessibility-aware interface design

Accessibility is part of product quality and should begin before visual polish.

WCAG 2.2 includes requirements relevant to interface design. Common controlled examples include:

- under SC 1.4.3, normal text generally needs at least **4.5:1** contrast, subject to stated exceptions;
- large text generally needs at least **3:1** contrast, subject to exceptions;
- under SC 1.4.4, text should be resizable up to **200%** without loss of content or functionality, subject to scope and technology conditions.

A good designer also considers:

- keyboard operation and logical focus order;
- visible focus states;
- headings and structure;
- labels and instructions;
- form errors and recovery;
- meaningful link/control names;
- reflow and zoom;
- content that does not rely on color alone;
- target size and spacing;
- motion and animation choices;
- time limits;
- status messages;
- alternative text/content responsibilities;
- screen-reader and assistive-technology behavior during implementation.

A design file cannot prove full WCAG conformance. Accessibility depends on implemented code, semantics, content, interaction behavior, device/browser combinations, assistive technology and applicable legal requirements.

Designers should identify risks and collaborate with qualified accessibility and engineering professionals rather than self-certify.

Inclusive and ethical user research

If user research is part of the role:

- obtain approval through the organization's research/privacy process;
- explain the purpose and expected participation honestly;
- collect only the data needed;
- avoid recording sensitive information without authorization;
- provide accessible participation options;
- avoid coercive or deceptive recruitment;
- protect recordings, notes and contact details;
- separate research evidence from stereotypes;
- document sample limitations;
- compensate participants according to approved procedures where applicable;
- do not publish identifiable participant information without authorization.

A designer is not automatically an institutional-review, legal, medical or privacy professional. Higher-risk research should involve the responsible experts.

Privacy and cybersecurity

Interface work may involve unpublished product information, user data, analytics, prototypes, research recordings, test credentials, screenshots and internal systems.

Practical safeguards include:

- use organization-approved storage and collaboration systems;
- enable MFA where supported;
- use unique credentials and approved password-management practices;
- verify unexpected file-share or login requests;
- limit prototype links to the intended audience;
- do not place real customer or employee data in public mock-ups;
- use synthetic or approved test data;
- remove unnecessary access when a project ends if that is your responsibility;
- report phishing, malware, accidental exposure or suspected compromise promptly;
- keep approved software and plugins updated;
- follow security review for authentication, payments, permissions and sensitive workflows.

CISA Secure Our World materials emphasize phishing awareness, strong passwords/password managers, MFA and software updates. Designers should apply those habits but should not invent security architecture or claim that an interface is secure without the responsible security/engineering review.

Responsible AI in interface work

AI can help with low-risk tasks when organizational policy allows it, such as:

- brainstorming alternative flows;
- generating fictional practice content;
- producing first-pass microcopy variants;
- explaining unfamiliar code;
- suggesting usability questions;
- organizing non-sensitive notes;
- exploring design-system documentation;
- generating synthetic examples for portfolio practice.

Human verification remains required.

Before using AI-assisted output, check:

- factual accuracy;
- accessibility implications;
- privacy and security;
- harmful or biased assumptions;
- licensing and provenance;
- brand and content standards;
- interaction logic;
- code safety and maintainability;
- whether disclosure is required;
- whether the organization's approved AI environment permits the data involved.

Do **not** upload confidential roadmaps, proprietary code, credentials, private user-research data, customer information or regulated/sensitive data to an unapproved AI service.

NIST's AI Risk Management Framework is voluntary guidance for managing AI risks, and its Generative AI Profile addresses risks particular to generative AI. The practical rule for this career is simple: **AI may assist the design process, but an authorized person must verify the final work, data handling, rights, accessibility, security implications and production readiness.**

Education and entry pathways — United States

O*NET currently places Web and Digital Interface Designers in a high-preparation occupational context, but actual hiring requirements vary. Some employers prefer a bachelor's degree; others accept an associate degree, certificate, bootcamp, strong portfolio, development experience or an equivalent combination.

A free-first sequence can be:

1. learn visual hierarchy, typography, spacing, responsive layout and basic interaction principles;
2. learn HTML and CSS concepts even if you plan to specialize in design;
3. learn accessibility fundamentals and keyboard/focus behavior;
4. create wireframes and prototypes for realistic tasks;
5. learn one modern interface-design/prototyping tool deeply enough to show components, states and responsive behavior;
6. build two or three portfolio cases with research assumptions, accessibility considerations and implementation notes;
7. practice collaborating with a developer or reviewing a coded implementation;
8. apply selectively to web, UI, UX/UI, interaction, digital-design or design-support roles aligned with your demonstrated level.

Funding, low-cost training and work-based learning — United States

CareerOneStop provides a Local Training Finder and state WIOA-eligible program directories. WIOA funding is not automatic; eligibility, approved programs and available services vary by state and individual. An American Job Center can explain local rules.

Useful search terms include:

- web design;
- UX design;
- UI design;
- interaction design;
- digital media;
- human-computer interaction;
- accessibility;
- front-end development;
- web development.

O*NET lists example apprenticeship titles related to this occupation, including Digital Accessibility Technician, User Experience Designer and

Web & Digital Interface Designer. A listed apprenticeship title does not guarantee that an active program exists near a learner. Verify current openings through Apprenticeship.gov and local workforce partners.

Canada pathway

Canada Job Bank uses **NOC 21233 — Web designers**. Its current description emphasizes visual presentation, layout and functionality of websites and related applications, ease of use and information access.

Job Bank says typical requirements include a bachelor's degree in computer science or graphic design, or completion of a college program in computer science, graphic arts or web design; experience as a graphic designer is usually required. Job Bank states that, according to its records, this occupation is **not regulated in Canada**.

Current national wage data:

- low: **C\$20.00/hour**;
- median: **C\$33.65/hour**;
- high: **C\$57.69/hour**.

Canada.ca provides training, student aid, loans/grants, skills-upgrading and work-experience locators. Verify provincial requirements, program quality, portfolio expectations, costs and eligibility before paying.

Colombia pathway

OCUPACOL identifies **CUOC 21662 — Diseñadores de imágenes computarizadas e interacción digital**. Its functions include designing digital content for applications, proposing intuitive and adaptable interfaces, creating prototypes, integrating user needs with functional requirements, planning information architecture and interaction flow, wireframing and testing.

Listed job names include UX designer, UI designer, interaction designer, web designer, digital designer, multimedia/web designer and front-end designer, among other related titles.

SENA

SENA Betowa currently lists **Diseño web con Adobe Dreamweaver**, a complementary virtual program totaling **40 hours**. It includes website/page configuration, CSS, multimedia integration and publication. The presence of this program is useful evidence of a current SENA web-design learning option, but **Dreamweaver is not treated in this guide as a required modern industry tool**. Learners should compare the current course content with the skills in real job postings before relying on it as their main preparation.

Betowa also surfaces **Diseño de sitios web** as an **80-hour** complementary course in current listings. Availability, municipality, modality, cohort and seats change.

Use Betowa as the current enrollment authority.

Latin America and Caribbean pathway

OIT/Cinterfor maintains country- and institution-level information for vocational-training organizations across Latin America and the Caribbean. Use it to identify the relevant national institution, then verify current training directly with that institution.

A regional directory does not guarantee that a particular UI/UX or web-design course, scholarship or subsidy is currently available.

United States employment outlook

O*NET's current occupation data report **128,900** people employed in 2024, projected growth of **7% or higher** for 2024–2034 (classified as much faster than average), and **9,100** projected openings over 2024–2034. These are occupation-level projections, not a guarantee that any individual learner will obtain a job.

Earnings research

United States — official occupation benchmark

O*NET/BLS 2025 national data for Web and Digital Interface Designers:

- 10th percentile: **\$53,750/year** or **\$25.84/hour**;
- 25th percentile: **\$73,290/year** or **\$35.24/hour**;
- median: **\$104,000/year** or **\$50.00/hour**;
- 75th percentile: **\$158,820/year** or **\$76.36/hour**;
- 90th percentile: **\$201,550/year** or **\$96.90/hour**.

These figures cover the broader occupation and should not be presented as entry-level guarantees.

United States — current non-government Web Designer estimate

Indeed reported approximately **\$26.94/hour** average base salary for the narrower title **Web Designer**, with a displayed range of about **\$13.39–\$54.17/hour**, based on **575** salaries from job postings in the preceding **36 months**, updated **August 2, 2026**.

This estimate is useful market context but is not directly interchangeable with the broader O*NET occupation.

Colombia — current non-government Web Designer estimate

Indeed Colombia reported approximately **COP 1,834,105/month** for **Diseñador/a web**, based on **30** published salaries and updated **June 9, 2026**.

This is a small, title-specific market sample. It is not an official CUOC wage benchmark and should not be generalized to all UX/UI, digital-product or front-end design roles.

Build an ethical portfolio

A portfolio should show how you think, not only attractive screens.

For each case study, explain:

- the problem and intended user;
- your role;
- constraints;
- evidence or assumptions;
- information architecture or flow;
- wireframes/prototypes;
- important interaction states;
- accessibility considerations;
- privacy/security considerations if relevant;
- feedback/testing;
- what changed;
- implementation/handoff notes;
- final outcome or what you would test next.

For self-directed work, use fictional brands and synthetic data or properly licensed public material. Never expose confidential employer/client work or real private user data simply to strengthen a portfolio.

Do not invent research participants, usability results, conversion improvements or business metrics. Clearly label hypothetical or practice work.

Practical portfolio projects

Good starter projects can include:

- an accessible appointment-booking flow;
- a responsive public-information website;
- an account settings page with error and recovery states;
- a simple e-commerce product and checkout concept using fictional data;

- a dashboard with clear filtering and empty/loading states;
- a redesign of a public-domain or fictional service, with the original problems and assumptions documented.

One thoughtful, tested case can be stronger than many decorative screens.

30-day action plan

Week 1 — Foundations

- Study hierarchy, typography, spacing, responsive layout and basic interaction principles.
- Learn the difference between UI, UX, web design and front-end implementation.
- Review WCAG 2.2 fundamentals for contrast, keyboard/focus, labels and zoom/reflow.
- Recreate one simple interface using components and consistent spacing.

Week 2 — Flow and prototype

- Choose one realistic user problem.
- Write the user's main task and success condition.
- Map the flow, including error and recovery states.
- Create low-fidelity wireframes before visual polish.
- Build an interactive prototype.

Week 3 — Evidence and accessibility

- Conduct a small ethical usability review with permission, or use a structured expert review if participants are unavailable.
- Record what you actually observed; do not invent findings.
- Run contrast and keyboard/focus checks against the design and implementation assumptions.
- Revise the prototype based on evidence.

Week 4 — Portfolio and job readiness

- Turn the project into a short case study.
- Research at least 10 real target postings.
- Build a skill-gap table from repeated requirements.
- Update your résumé and portfolio using only skills you can demonstrate.
- Apply selectively while continuing one specific learning gap.

Before handing a design to development

Check:

- correct approved flow and version;

- layouts for required breakpoints;
- component names and states;
- hover/focus/selected/disabled/error/loading/success behavior where relevant;
- real or approved content rather than unexplained placeholder text;
- labels and instructions are clear;
- contrast and accessibility considerations reviewed;
- keyboard/focus expectations documented;
- assets and icons supplied correctly;
- spacing/tokens/style rules are consistent;
- edge cases and validation are addressed;
- no confidential or real sensitive data are exposed;
- engineering/security/privacy questions are identified rather than guessed;
- approval and handoff channel are clear.

Questions to ask before paying for training

- Does the curriculum teach current interface and responsive-design practice?
- Does it cover accessibility rather than treating it as optional?
- Will I build portfolio work I am legally allowed to show?
- Does it include interaction states and prototypes, not only static graphics?
- Does it teach basic HTML/CSS concepts or collaboration with developers?
- Are instructors currently experienced in the field?
- Are software costs included or extra?
- Is there supervised work, internship, co-op or employer-connected practice?
- Is funding actually approved for me, or is the provider only saying it may be eligible?
- What do recent graduates produce and what evidence supports employment claims?

Sources to verify before important decisions

- O*NET Web and Digital Interface Designers: [Source link](#)
- O*NET/BLS national wages: [Source link](#)
- CareerOneStop WIOA finder: [Source link](#)
- CareerOneStop training finder: [Source link](#)
- Canada Job Bank summary: [Source link](#)
- Canada Job Bank requirements: [Source link](#)
- Canada Job Bank wages: [Source link](#)

- Canada training gateway: [Source link](#)
- Colombia OCUPACOL CUOC 21662: [Source link](#)
- SENA Betowa web-design offer: [Source link](#)
- SENA Betowa catalogue: <https://betowa.sena.edu.co/>
- OIT/Cinterfor country/institution locator: <https://www.oitcinterfor.org/statsfp/paises>
- WCAG 2.2: <https://www.w3.org/TR/WCAG22/>
- CISA Secure Our World: <https://www.cisa.gov/secure-our-world>
- NIST AI RMF: [Source link](#)
- NIST Generative AI Profile: [Source link](#)
- Indeed U.S. Web Designer market estimate: [Source link](#)
- Indeed Colombia Web Designer market estimate: [Source link](#)

Important notice

This guide provides general educational and career-planning information. It does not guarantee employment, income, admission, funding, licensing, certification, apprenticeship placement, freelance work, client acquisition, promotion or any other result. Requirements, compensation, tools, standards and opportunities change by employer, jurisdiction and time.

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